

Giacomo Manzoni

giacomo.science@gmail.com | [in](#) | [🌐](#)

EDUCATION

ETH Zürich

M.Sc. in Robotics, Systems and Control. GPA 5.7 out of 6

Zurich, CH

Sep 2021 – Jun 2024

Politecnico di Milano

B.Sc. in Automation Engineering. Maximum grade. Top 5%

Milano, IT

Sep 2018 – Jul 2021

EXPERIENCE

Robotics Engineer - Robot Learning

Hexagon Robotics

Zurich, CH

Sept 2024 – present

Design and development of Imitation Learning framework for dexterous manipulation. From early stage prototyping to product launch.

- Python • Development and integration of Imitation Learning models. Outperformed the state of the art
- Pytorch in success rate and reliability
- ROS2 • Implementation of fully integrated pipeline. Teleoperation, data collection, augmentation, train-
- C++ ing, evaluation and inference
- ONNX • Both on real hardware and in an Isaac-sim. Training locally and in cloud
- Lead efforts to deliver a world-first live humanoid manipulation demo, on stage

R&D Internship

Leica Geosystems, part of Hexagon

Zurich, CH

Feb 2023 – Jul 2023

- C++ • Design and development of Path Planning algorithms for [BLK2FLY](#)
- ROS • Exploration and 3D mapping (LIDAR) of unknown environments
- Python

SELECTED PROJECTS

Master's Thesis - Imitation Learning for Dexterous Manipulation

SRL

Zurich, CH

Oct 2023 – Jun 2024

Embodiment agnostic pre-training of foundation model for dexterous manipulation.

- Python • Designed pipeline for data collection on real hardware: camera calibration - monocular 6dof
- PyTorch hand pose estimation - distributed ROS - teleoperation
- Implemented framework for visual encoder finetuning
- Trained various generative models for the manipulation policy
- Work done as student researcher supervised by Elvis Nava (now [Mimic Robotics](#))

Motion-Primitives Planning for Legged Robots

RSL

Zurich, CH

Sep 2022 – Feb 2023

- C++ • Integrating motion primitives in local path-planning for legged robots
- Python • Collaboration between ETH Zurich and the University of Oxford

SKILLS

Python	PyTorch	Linux	Git	OpenCV
ROS2	IsaacLab	Deep Learning	Imitation Learning	Path Planning

LANGUAGES

Italian (native)	German (basic)	English (proficient)
------------------	----------------	----------------------